

1.1 INTRODUCTION

Project management is the application of knowledge, skills, tools and techniques to project activities in order to meet the project requirements. Project management is the process of planning, organizing and executing the activities to realize a brilliant idea into a product, service or deliverable (Project Management Institute, 2026). In Project Management, there are several ways to define project success. This includes the project met scope, time, and cost goals. Then, results of the project also met its main objective, such as making or saving a certain amount of money, providing a good return on investment, or simply making the sponsors happy.

In the context of information technology, project management become even more critical because IT projects often involve highly technical works, large teams and more complex stakeholder needs. The failure to properly managed an IT project can leads to cost overruns, delayed delivery, and systems that does not meet user requirements. Studies has shown that a large percentage of IT projects fails due to poor planning and lack of proper project management methodology. Recent trends affecting IT project management include globalization, outsourcing, virtual teams, and agile project management.

There are five process groups that involves project management which is Initiating, Planning, Executing, Monitoring and Controlling, and Closing. The project starts with the initiating phase, where the project is defined and approval is obtained before the project can begin. During the planning phase, the project scope, schedule, budget, and resources are prepared to guide the project execution. The executing phase involves carrying out the project activities and usually require the most time and resources. In this phase, the project manager must demonstrate leadership to ensure that the project team and stakeholders are working together.

The monitoring and controlling phase take place throughout the project life cycle to monitor project progress, compare the actual performance with the project plan and make correction if needed. Lastly, the closing phase involves getting stakeholder acceptance, documenting lessons learned, and formally ending the project. These five process groups help organizations to manage complex IT projects more effectively and reduce the risk of delays, cost overruns, and project failure. These process groups are used to ensure that a project is successful. Project integration management is needed to effectively coordinate all these activities.

1.2 PROJECT INTEGRATION MANAGEMENT

Project management is a very complex area and requires a clear and structured approach to ensure the project is successful. Integration is a key element in every project because it is ensuring that every component of the project collaborates to achieve the desired result. Project Integration Management, the process of bringing together and integrating all aspects of a project. Project Integration Management includes the processes and activities to identify, define, combine, unify, and coordinate the various processes and project management activities within the Project Management Process Groups. This will involve managing tasks, resources, stakeholders and other aspects of a project to ensure that it is all linked up. It also require dealing with any conflicts among the project components, resolving competing demands, and managing the use of resources.

In the simplest of terms, project integration management ensures that all work performed in the project is adequately coordinated and can run smoothly. In simpler words, integration management is about making sure that all the different parts of a project are working together in a coordinated and consistent manner. Without proper integration, each team may work independently without align with the overall project goals, which can leads to miscommunication, duplication of efforts, and ultimately, project failure (Artem Gurnov, 2024).

Project Integration Management can be described by seven key processes after understanding the concept of integration. It begins with Develop Project Charter, which is the project's formal approval and outlines some of the initial details including objectives, stakeholders, and the project manager's authority. Develop Project Management Plan is the next step, and brings together all subsidiary plans from each of the knowledge areas into one single guiding document. The Direct and Manage Project Work process is the actual execution of the project activities to deliver the project to meet the plan.

Moreover, Manage Project Knowledge enables the sharing of useful knowledge and the capturing of lessons learned throughout the project. Then, the Monitor and Control Project Work process is used to measure progress, compare performance to the project plan and maintain alignment with project goals. After that, Integrated Change Control processes all change requests, considers and approves required changes to ensure project control. Lastly, Close Project or Phase is the final activity that brings the project or phase to a close by completing project activities and reporting project results. These processes manage the coordination, control, and alignment of the various project components so that the project is completed successfully.

1.3 PROJECT INTEGRATION MANAGEMENT FOR TENDER PERKHIDMATAN PEMBANGUNAN SISTEM PENGURUSAN MAKLUMAT POLITEKNIK (SPMP)

Project Integration Management for the SPMP Tender is a project management process that coordinates links all of the other project management processes to ensure that the development is accomplished successfully from the beginning of the project, until the project is completed. This project involves the development of a number of components including software development, database management, ICT security and data management, which need to be integrated to a full and working system, in line with the tender issued by Jabatan Pendidikan Politeknik dan Kolej Komuniti (JPPKK). The process of integration starts with the project charter, which is an authorization document and sets the project's broad purpose, with budget of RM4,140,000.00.

ELEMENTS	SPMP PROJECT DETAILS
Project Title	Perkhidmatan Pembangunan Sistem Pengurusan Maklumat Politeknik (SPMP)
Sponsoring Agency	Jabatan Pendidikan Politeknik dan Kolej Komuniti (JPPKK)
Project Purpose	To develop an integrated information management system consisting of 14 modules: i. Hal Ehwal Pelajar (HEP) ii. Pendaftaran Kursus (termasuk Ko-Kurikulum) iii. Pengurusan Pengguna (termasuk Kod Selenggara) iv. Penilaian v. Fail Rancangan Mengajar (FRP) vi. Penasihat Akademik vii. Pemantauan PDP viii. Kurikulum ix. Latihan Industri x. Pengurusan Projek Pelajar xi. Kamsis xii. Kaunselor xiii. Konvokesyen xiv. Alumni
Planned Activities	ICT 01 Analysis, ICT 02 Design, ICT 03 Coding, ICT 04 Testing, ICT 05 Implementation, ICT 06 Latihan Pengguna Sistem, ICT 07 Latihan Teknikal ICT 08

	Pengurusan Perubahan, ICT 09 Keselamatan Sistem and ICT 10 Perkhidmatan Pengoperasian Dan Sokongan Teknikal
Estimated Budget	RM4,140,000.00
Duration	36 months

Next, Develop Management Plans. It is a comprehensive project management plan would need to be developed that integrates all subsidiary plans. Then, Direct and Manage Project Work. For the SPMP project, this means executing the planned activities across all ten ICT phases that stated in the specification. This includes ICT 01 Analysis, ICT 02 Design, ICT 03 Coding, ICT 04 Testing, ICT 05 Implementation, ICT 06 Latihan Pengguna Sistem, ICT 07 Latihan Teknikal ICT 08 Pengurusan Perubahan, ICT 09 Keselamatan Sistem and ICT 10 Perkhidmatan Pengoperasian Dan Sokongan Teknikal. All of the planned activities are supposed to be for local resources, and it is ensuring that all these activities are executed according to plan.

Other than that, Manage Project Knowledge. SPMP system involves technical knowledge about polytechnic operations, academic processes, and ICT development. Then, for Close Project or Phase. The SPMP project has a clear closing phase, with the Final Acceptance letter. Once the letter is issued, it signifies that the system has met all requirements and the project can be formally closed.

1.4 IMPORTANCE OF INTEGRATION MANAGEMENT

Project Integration Management is the process of bringing together and combining all aspects of a project in a coordinated way. It includes processes and activities to identify, define, combine, and coordinate the various project management processes within the Project Management Process Groups. Without strong integration management, the SPMP project could easily fall into common IT project failure traps. For examples, if the analysis team produced any documents that does not properly reflect what the design team needed, the entire project development would be affected.

Integration management ensures that all the processes is properly connected and aligned. It helps improve the efficiency of a project by ensure that resources are used in the

best way. The project progress must be monitored, and it is needed to find areas that need improvement. This helps to avoid each person doing the same work twice, reduce wasted time to increase productivity. Besides that, Project Integration Management helps increase responsibility among team members. When the project goals, roles, and responsibilities are clearly explained, this reduce confusion and helps team members communicate better with each other. As team members become more responsible for their own work, problems can be identify earlier before they become bigger issues. Therefore, good teamwork and clear responsibilities can improve the overall project performance (Adobe for Business Team, 2025).

In conclusion, Project Integration Management serve as the backbone of the entire project management framework. For a complex system like SPMP that involves multiple stakeholders, and a high budget of RM4,140,000.00, integration management is absolutely essential for the project to be deliver successfully. Integration management plays an important role in making sure every process is connected and aligned with the overall project goals.

4.1 - INTRO CHAPTER SUMMARY COST

Project Cost Management is one of the important knowledge areas in project management because it helps ensure that a project is completed within the approved budget. It involves planning, estimating, budgeting and controlling project costs throughout the project lifecycle. Since many IT projects face budget problems, project managers need to understand the basic principles of cost management so they can make better financial decisions and avoid unnecessary spendings. Cost is generally define as a resource that is sacrificed or given up in order to achieve a specific project objective, and it is usually measured in monetary value such as Ringgit Malaysia in this context.

There are four main processes in Project Cost Management, which is Plan Cost Management, Estimate Costs, Determine Budget, and Control Costs. Plan Cost Management involves determining the policies, procedures and documentation that will be used to plan, manage and controlling project cost. Next, Estimate Costs is the process of calculating expected expenses for each activity. Then, Determine Budget involves allocating the overall cost estimate to individual work items in order to establish a cost baseline that can be use to measure project performance. Lastly, Control Costs involves monitoring the project's cost performance, managing changes to the cost baseline, and informing stakeholders of any authorized changes that may affects the budget (Atlassian, 2026).

These processes help project managers to estimate the project costs, prepare a realistic budget, monitor project expenses, and control any changes that may affect the project budget. By applying these processes correctly, the project has a higher chance to be completed within the planned cost while still meeting the project objectives.

The SPMP project has a total indicative budget of RM4,140,000.00 for a contract duration of 36 months. This is a significantly large budget, which is reflects the complexity and scale of the system being developed, especially since it consist of 14 functional modules to be used by polytechnics and community colleges nationwide. Managing this budget effectively is one of the most critical responsibilities of the project team.

ELEMENTS	COST DETAILS
Total Budget	RM4,140,000.00
Duration	36 months
Major Cost	ICT 03 – Coding & Development (14 modules)

Other	BRS, SRS, SDS workshop sessions; Training (5 sessions); Operational & Technical Support
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The cost of the SPMP project would be distributed across the ten ICT phases. Some phases, such as system coding and development (ICT 03), which involve building 14 functional modules, would likely represents the largest portion of the budget due to the total manpower required for analysis, coding and testing of each modules. Other cost components include the workshop sessions for Business Requirements Specification (BRS), System Requirements Specification (SRS), System Design Specification (SDS), and the cost of training (5 sessions) which is covering both user training and technical training, as well as the cost involve in providing operational and technical support after the system is implement.

This particular requirement has an important impact on cost planning because it is needed to need to estimate all possible costs from the beginning, even before the detailed analysis phase begins. This include manpower cost, training cost, and post-implementation support cost, which all of it need to be accounted for within the indicative budget figure. If the cost is not estimate correctly at this early stage, it may cause financial problems during the project, especially since the budget is not easily adjusted once the contract has been agreed upon. Due to that, good Project Cost Management is important to avoid budget issues and help the SPMP project achieve its objectives within the planned budget of RM4,140,000.00, across the full 36-month contract duration.

CONCLUSION

In conclusion, project management is an important process in ensuring that project can be completed successfully. Several project management knowledge areas have been discussed including Project Integration Management, Project Planning Management, Project Scope Management, Project Cost Management, Project Human Resource Management, Project Communication Management and Project Risk Management.

Based on the Tender Perkhidmatan Pembangunan Sistem Pengurusan Maklumat Politeknik (SPMP), it can be seen that managing a large IT project is not only about developing a system, but it involves proper planning, coordination and continuous monitoring. The project consists of ten ICT phases, fourteen functional modules and an indicative budget of RM4,140,000.00. Because of that, every project activity need to be planned carefully so that the project can be completed according with the project requirements and within the planned schedule and budget.

Besides that, every knowledge area supports each other throughout the projects. Integration Management helps coordinate all project activities while Planning Management provides a clear direction before the project starts. Scope Management makes sure the project deliverables are clearly defined, Cost Management controls the project budget and Human Resource Management ensures the project team works effectively. Communication Management helps information being shared among stakeholders, Risk Management helps identify and respond to possible project risks before they become more serious. If one of these areas are not managed properly, it may affect the overall success of the project.

The SPMP project shows that good project management is very important in government IT projects because they usually involve large budgets, many users, and long project durations. Proper management can help reduce misunderstandings, avoid unnecessary delays, improve teamwork, and ensure that project resources are used effectively. By applying the project management processes correctly, as proves, an effective project management is one of the key factors that contribute to the success of the SPMP project and other future IT projects as well.

Adobe for Business Team. (April 14, 2025). *Project integration management, definition, process, and steps*. Adobe.Com. <https://business.adobe.com/blog/basics/integration-management>

Artem Gurnov. (2024, September 10). *What Is Project Integration Management?* Wrike.Com; Wrike. <https://www.wrike.com/project-management-guide/faq/what-is-project-integration-management/>

Atlassian. (2026). *Project Cost Management: Processes & Best Practices*. Atlassian.Com. <https://www.atlassian.com/work-management/project-management/project-cost-management>

Project Management Institute.(2026). *What Is Project Management*.Pmi.Org. <https://www.pmi.org/about/what-is-project-management>